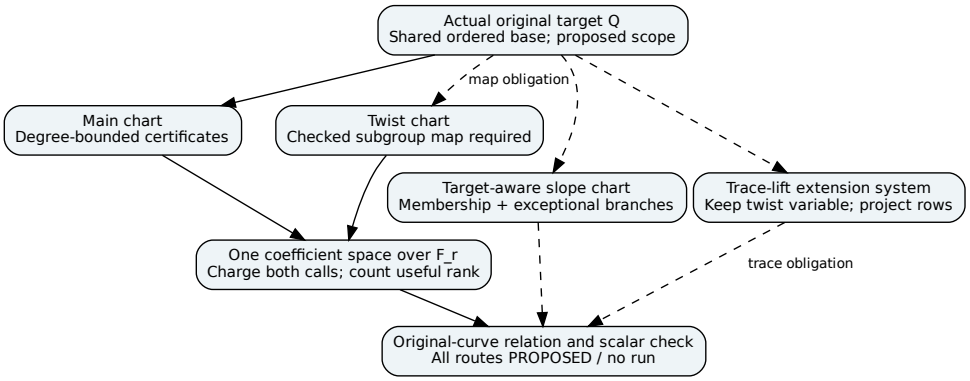
Quadratic twists: three proposed research routes

2026-11-05 Follow-up to merged PR2381

Status. These are design hypotheses, with PCC statement standing `unverified`. There is no new scientific run, theorem, native algorithm, measurement or gain in this proposal record. [pcc:unverified]



1. Shared degree-limited collector

Use an actual scalar-preserving subgroup map with an explicit inverse, and transport one ordered factor base. For actual unknown Q, the two models supply admitted coefficient rows in the SAME space over $\mathbf{F}_r$. The curve field and relation field are different objects.

Let $R_E(D, Q)$ and $R_T(D, \phi(Q))$ be degree-limited row sets. The proposed discriminant is extra useful rank of their combined span relative to either standalone span, under the SAME total-work prefix. This is a hypothesis, not a measured gain. [pcc:unverified]

A complete-or-unknown oracle may release the same full set or no rows. Test this guard explicitly before interpreting per-query complementarity. Across query prefixes, use frozen main-only, twist-only, alternating and dual-model schedules. Charge both dual calls, all failures, setup, routing, maps, matrices and row work. A post-hoc best-chart oracle is diagnostic only.

Actual source recovery requires original-valid rows, legitimate public base logs and an original-point scalar check. Rank-deficient setup blocks every dependent target. No label oracle or unlimited fallback is admissible.

PCC statement: `PCC-20261105-1f629a`. All scientific assertions remain unverified. The machine plan and detailed acceptance conditions are adjacent.

2. Full-target chord and membership chart

For an elliptic short model $y^2 = x^3 + Ax + B$ in characteristic other than 2 or 3, start with a nonidentity actual target $R = (a, b)$. Candidate lines through $-R$:

$$y = \lambda(x - a) - b.$$

The proposed residual intersection quadratic is

$$q_\lambda(X) = X^2 - sX + p, \qquad s = \lambda^2 - a,$$

$$p = a\lambda^2 + 2b\lambda + a^2 + A.$$

The first proof target is the curve-minus-line factorization by $(X - a)q_\lambda(X)$, using the actual target equation. [pcc:unverified]

For a three-abscissa base,

$$F_B(X) = X^3 - c_1X^2 + c_2X - c_3,$$

the candidate remainder modulo q_λ is $UX + V$, with

$$U = s^2 - p - c_1s + c_2, \qquad V = (c_1 - s)p - c_3.$$

Distinct-root membership would require $U = V = 0$ and actual field/point lifts. The proposed cancellation $V + aU$ gives a quartic/cubic gcd starting chart. These are unverified formulas awaiting a faithful pinned Lean proof and source-bound identity controls. No constant-degree claim for growing bases is made. [pcc:unverified]

Keep all branches. A square-free base polynomial need not contain a repeated-root quadratic. Keep diagonal/tangent, origin, vertical-line, coincident-target, splitting, denominator and sign cases explicitly. Establish coverage before admitting a reduced system.

Offer the same chart to the source, twist and transported controls. Record matrix degree, expanded original-input witness degree, weighted/pulled-back degree, extraction and group work separately. Compare against an applicable source-side univariate reference. Pair tables cannot supply candidate stopping.

PCC statement: `PCC-20261105-585d85`. Background: Semaev; Kosters–Yeo, sections 2 and 4–5. Input and first-fall degrees do not by themselves supply solving-cost evidence.

3. Twist variable and source trace

Let $K = \mathbf{F}_{q^2}$ and $\tau(X) = X + \mathrm{Frob}_q(X)$. Use an actual additive twist embedding ψ onto the Frobenius-minus subgroup. Let B_i, Q lie in an explicit cyclic subgroup of odd order r, and choose public h with $2h \equiv 1 \pmod{r}$. Candidate lifts are

$$C_i = \psi(T_i), \quad A_i = [h]B_i + C_i, \quad A_Q = [h]Q.$$

The target equivalence is

$$\big(\exists T,\ \sum_i c_i A_i = A_Q + \psi(T)\big) \quad \Longleftrightarrow \quad \sum_i c_i B_i = Q.$$

Apply trace in one direction and construct the residual twist point in the other. This is an unverified theorem target on actual groups, not a completed conditional substitute or a global rational-group isomorphism. [pcc:unverified]

Retain T as an auxiliary variable and test intermediate equation demand. Charge the extension variables, field arithmetic and maps; retain expanded source witnesses. Compare against a source auxiliary-variable encoding and collapse witnesses projecting to the same coefficient row before rank. Multiple twist witnesses alone are not multiple useful relations.

PCC statement: `PCC-20261105-cfd582`. Background: Gorla–Massierer, trace and relation construction; Galbraith–Granger–Merz–Petit, Frobenius-compatible bases.

First scope. Collector and chord completeness preflight first; trace construction queued. Draft cards are not runs or task leases. Before new data: current source/identity pins, native producer and separate arithmetic auditor, faithful Lean, canonical frozen protocol, controls and report. Use at most two existing local workers, 600-second stages and one focused window no longer than 3600 seconds. No paid provisioning.

Concrete launch dependency. Prior checkpoint approval covers only the old branch. Additional scope is required for

```
s3://setec-astronomy-artifacts/checkpoints/crypto/
codex__twist-collector-and-slope/
```

Use the current unmodified official gate after approval; no gate/upload is performed now. The merged branch and old evidence remain unchanged.

Visual scope. This diagram records proposed routes only. There are no new numeric results, so canonical quantitative graphs and benchmarks remain unchanged. See adjacent README, IDEAS, PLAN and LAUNCH_GATES for source links, PCC status, controls and the exact open dependencies. [pcc:unverified]